

CURRICULUM VITAE – YUNFEI MA (U.S. PERMANENT RESIDENT)

SENIOR ENGINEERING MANAGER, MOBILE AND EDGE NETWORKING

PERSONAL INFORMATION

Yunfei Ma
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CURRENT POSITION

Senior Engineering Manager
Uber Technologies, Inc.
Leading Uber's Mobile and Edge Networking organization
Sunnyvale, CA

PROJECTS & ACHIEVEMENTS (SELECTED)

Uber SkyEdge Evolution: I led the re-architecture of Uber's anti-DDoS infrastructure and the cloud migration and evolution of Uber's edge and mobile networking infrastructure. This project saved Uber \$39M over three years and contributed a \$200M uplift to mobility gross bookings over the last two years.

Uber Anti-DDoS Leadership: Recognized as a 2024 **Uber Re-imagine Award** finalist (Uber's highest company award) for driving Uber's anti-DDoS strategy, leading the Cloudflare migration, and re-architecting a layered defense model.

Large-scale Deployment: (1) I led the design and development of *XLINK*, the industry's first widely deployed multi-path QUIC stack. *XLINK* solves the performance bottleneck of traditional multipath transport protocols through a cross-layer approach. The technology is currently deployed in Alibaba's Taobao, Eleme, AliCloud, Dingtalk, and many other services, serving hundreds of millions of users. (2) I led the design and development of *GSO-Simulcast*, a globally controlled stream orchestrator deployed at full scale in Alibaba's Dingtalk video conferencing service, serving hundreds of millions of users. The overall user satisfaction score improved by 7% after the deployment of *GSO-Simulcast*.

New Business Incubation: I initiated and led the development of *Cellfusion*, a new connectivity-as-a-service product in the AliCloud IoT family, tailored for self-driving customers. *Cellfusion* is a novel hardware-software system that aggregates multiple heterogeneous cellular networks to achieve stable and high-quality vehicle-to-cloud communication.

Standardization: I am a co-author of the IETF Multipath QUIC draft. This draft was adopted as a working group (WG) draft at IETF 112 and is now progressing toward RFC publication. Since the introduction of *XLINK*, many companies, including Ericsson, Akamai, and Cloudflare, have begun implementing multipath QUIC.

Leadership: At Alibaba, I led an R&D team building cutting-edge networking technologies. I served as the overall technical lead for Alibaba's predictable wide-area network (PredWAN) infrastructure, which combines QUIC and SRv6. I led R&D efforts across QUIC-aware proxying, traffic engineering,

multi-cloud overlay networking, network monitoring, and CDN performance optimization.

WORK EXPERIENCE	Uber Technologies Sunnyvale, CA, Sept. 2023–present Sr. Staff Engineer → Sr. Staff & TLM → Sr. Engineering Manager I lead Uber’s Mobile and Edge Networking organization, managing 22 engineers and serving 200M monthly active users across Uber’s mobile and web platforms. The Edge Networking team manages all Layer 7 ingress traffic into Uber, while the Mobile Networking team owns the mobile networking stack, client libraries, and observability infrastructure across all Uber apps on iOS and Android. My organization also manages Uber’s CDN infrastructure and anti-DDoS infrastructure. Alibaba Group (U.S.) Sunnyvale, CA, Sept. 2018–Aug. 2023 Senior Engineer → Staff Engineer → Sr. Engineering Manager I led a team building cutting-edge technologies in mobile and wide-area networking. Our technologies were deployed in Alibaba’s core services, including Taobao, Dingtalk, Eleme, and AliCloud, serving hundreds of millions of users worldwide. I served as the technical lead for Alibaba’s PredWAN global infrastructure, spanning transport protocols, traffic engineering, proxy acceleration, network quality monitoring, multi-cloud overlay networking, CDN congestion control, and video conferencing optimization. I was a committee member of Alibaba Mobile Technology Group and Alibaba Real-time Communication Technology Group. MIT Media Lab Cambridge, MA, Sept. 2016–Aug. 2018 Postdoctoral Associate My research focused on Internet-of-Things (IoT)-based human-machine interfaces. I used tools from computer networks, signal processing, machine learning, and hardware design to uncover, analyze, and engineer the networks of natural and man-made signals that fill our world in order to enhance our abilities in communication, sensing, and actuation.
EDUCATION	Ph.D. in Electrical and Computer Engineering Cornell University (2010–2016). GPA: 4.21/4.3 B.S. in Electrical Engineering University of Science and Technology of China (USTC) (2006–2010). GPA: 4.00/4.3
SELECTED HONORS AND AWARDS	Uber Re-imagine Award finalist, 2024 Keynote speaker at APNET’22 and APNET’21

Keynote speaker at MobiHoc'21
 MobiCom'21 Best Paper Runner-Up
 DAMO Award Individual finalist (the highest honor in DAMO Academy; Top 0.4%)
 Top-10 Breakthrough News story by RFID World, 2020
 Covered by IEEE Signal Processing Magazine Special Report, 2018
 ACM SIGCOMM research paper highlights, 2018
 China Young Scientist Computing Association Rising Star, 2018
 Qualcomm QualStar Award, 2015
 International Microwave Symposium Best Student Paper Award, 2015
 Irwin and Joan Jacobs Scholar, 2011
 National Scholarship (Top 2%), Ministry of Education, China, 2009

IETF DRAFTS

Y. Liu, Y. Ma, Q. D. Coninck, O. Bonaventure, C. Huitema, M. Kuhlewind, *Multipath Extension for QUIC*, IETF WG Draft, 2023. **Working group draft in the IETF QUIC WG.**
 Z. Zheng, Y. Ma, Y. Liu, M. Kuhlewind, *QUIC-enabled Service Differentiation for Traffic Engineering*, IETF Draft, 2023.
 F. Yang, Y. Liu, Y. Ma, *A Configurable Retransmission Extension for HTTP/3 Datagrams*, IETF Draft, 2023.
Y. Ma, Y. Liu, C. Huitema, X. Yu, *An Advanced Scheduling Option for Multipath QUIC*, IETF Draft, 2023.

SELECTED PUBLICATIONS

Y. Ni, Z. Zheng, X. Lin, F. Gao, X. Zeng, Y. Liu, G. Yang, Y. Su, D. Cai, H. Liu, C. Xu, E. Zhai, Y. Ma *corresponding author*, *CellFusion: Multipath Vehicle-to-Cloud Video Streaming with Network Coding in the Wild*, ACM SIGCOMM 2023. **Cellfusion is a new AliCloud product designed for self-driving customers.**
 B. Wu, K. Qian, B. Li, Y. Ma, Q. Zhang, Z. Jiang, J. Zhao, E. Zhai, D. Cai, X. Jin, *XRON: A Hybrid Elastic Cloud Overlay Network for Video Conferencing at Planetary Scale*, ACM SIGCOMM 2023.
 X. Lin, Y. Ma *corresponding author*, J. Zhang, Y. Cui, J. Li, S. Bai, Z. Zhang, D. Cai, H. Liu, M. Zhang, *GSO-Simulcast: Global Stream Orchestration in Simulcast Video Conferencing*, ACM SIGCOMM 2022. **Deployed in Dingtalk, serving over 500 million users worldwide.**
 Z. Zheng, Y. Ma *corresponding author*, Y. Liu, F. Yang, Z. Li, Y. Zhang, J. Zhang, W. Shi, W. Chen, D. Li, Q. An, H. Hong, H. Liu, and M. Zhang, *XLINK: QoE-driven Multi-path QUIC Transport in Large-scale Video Services*, ACM SIGCOMM 2021. **Deployed in Taobao and successfully spun off into a cloud product.**
 X. Fan, L. Shangguan, S. Rupavatharam, Y. Zhang, J. Xiong, Y. Ma, and R. Howard, *HeadFi: Bringing Intelligence to All Headphones*, ACM MobiCom 2021. **Best paper runner-up.**
 R. Zhao, P. Wang, Y. Ma *corresponding author*, P. Zhang, H. Liu, X. Lin, X. Zhang, C. Xu, and M. Zhang, *NFC+: Breaking NFC Networking Limits through Resonance Engineering*, ACM SIGCOMM, 2020. **Selected as one of RFID World's Top-10 Breakthrough News stories.**
 X. Fan, L. Shangguan, R. Howard, Y. Zhang, Y. Peng, J. Xiong, Y. Ma, X.-Y. Li,

Towards Flexible Wireless Charging for Medical Implants Using Distributed Antenna System, ACM MobiCom, 2020.

Z. Luo, Q. Zhang, Y. Ma, M. Singh, and F. Adib, *3D backscatter localization for fine-grained robotics*, USENIX NSDI, 2019. *Featured on MIT News, RFID Journal, Engadget, ZDNet, and other media outlets*

U. Ha, Y. Ma, Z. Zhong, T. Hsu, and F. Adib, *Learning food quality and safety from wireless stickers*, ACM HotNets, 2018. *Story covered by CBS This Morning*

Y. Ma, Z. Luo, C. Steiger, G. Traverso and F. Adib, *Enabling deep-tissue net-working for miniature medical devices*, ACM SIGCOMM, 2018. (acceptance rate: 40/222 = 18%) *Featured on Technology Review, Boston Herald, Engadget, and other media outlets*

Y. Ma, N. Selby, and F. Adib, *Minding the billions: ultra-wideband localization in deployed RFID tags*, ACM MobiCom, 2017. (acceptance rate: 35/186 = 19%)

Y. Ma, N. Selby, and F. Adib, *Drone relays for battery-free networks*, ACM SIGCOMM, 2017. (acceptance rate: 36/250 = 14%) *Featured as a Spotlight on the MIT homepage, IEEE Spectrum, The Verge, Sina, Sohu, and other media outlets*

Y. Ma, N. Selby, M. Singh, and F. Adib, *Demo: fine-grained RFID localization via ultra-wideband emulation*, ACM SIGCOMM, 2017.

Y. Ma, X. Hui, and E. C. Kan, *3D real-time indoor localization via broadband nonlinear backscatter in passive devices with centimeter precision*, ACM MobiCom, 2016. (acceptance rate: 32/226 = 14%)

MEDIA COVERAGE	CBS This Morning, The Verge, IEEE Spectrum, MIT News (front page), International Business Times, Boston Herald, Technology Review, Digital Trends, Geek, Inverse, Engadget, UPI, RFID Journal, Cornell News, ScienceDaily, Sina (Chinese), Sohu (Chinese), IT Home (Chinese), The Drive, Material Handling and Logistics, RF Global Net, Computer Welt (Austria), Electronics360, Quartz, New Atlas, Design Products and Applications.
INVITED REVIEWER FOR	IEEE Transactions on Mobile Computing IEEE/ACM Transactions on Networking IEEE Consumer Electronics Magazine
OTHER SERVICE	USENIX NSDI Technical Program Committee 2027 ACM MobiCom Technical Program Committee 2025 ACM SIGCOMM Technical Program Committee 2024 ACM HotMobile Technical Program Committee 2024 IEEE GlobeCom Technical Program Committee 2021 IEEE INFOCOM Technical Program Committee 2020 IEEE INFOCOM Technical Program Committee 2019 ACM CoNEXT Technical Program Committee 2018 ACM MobiCom Workshop Technical Program Committee 2018 ACM Mobisys Workshop Technical Program Committee 2018